

● EASY

● ADVANCED MATH

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# Advanced Math

# 04

10 questions · ~15 min

**Q1**

ADVANCED MATH

Which expression is equivalent to  $15w^2 + 8w$ ?

- A.  $w15w + 8$
- B.  $8w15w + 1$
- C.  $15w^28w + 1$
- D.  $23w^2 + w$

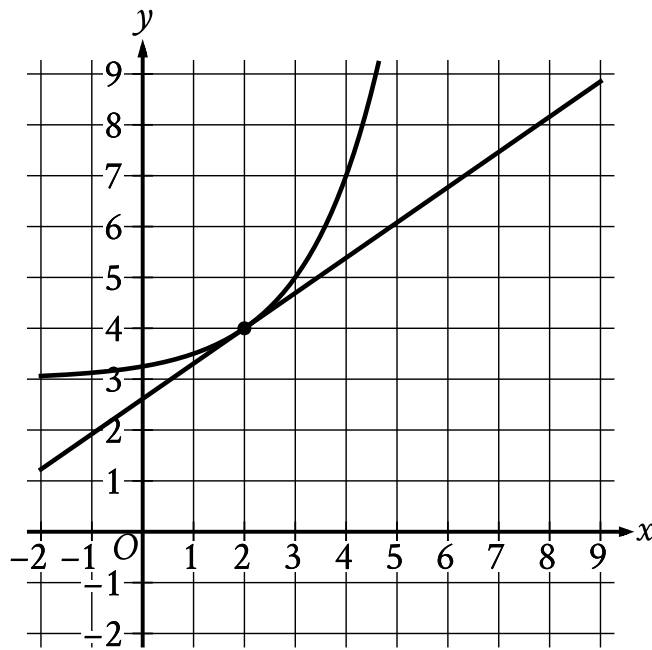
The function  $f$  is defined by  $fx = \frac{16}{x}$ . What is the value of  $fx$  when  $x = 17$ ?

A.  $\frac{16}{17}$

B.  $\frac{17}{16}$

C. 16

D. 17



- For the curve in the system:
  - Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1.
    - In quadrant 2, the curve trends up gradually to point (0 comma 3.25).
    - In quadrant 1, the curve trends up sharply.
  - The curve passes through the following points:
    - (1 comma 3.5)
    - (2 comma 4)
    - (4 comma 7)
- For the line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (2 comma 4)
    - (7 comma 7.5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution  $x, y$  to this system?

A. 0,0

B. 0,2

C. 2,4

D. 4,0

Which of the following expressions is equivalent to  $2(ab - 3) + 2$  ?

A.  $2ab - 1$

B.  $2ab - 4$

C.  $2ab - 5$

D.  $2ab - 8$

The function  $f(x) = 200,000(1.21)^x$  gives a company's predicted annual revenue, in dollars,  $x$  years after the company started selling light bulbs online, where  $0 < x \leq 10$ . What is the best interpretation of the statement " $f(5)$  is approximately equal to 518,748" in this context?

- A. 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately 518,748 dollars.
- B. 5 years after the company started selling light bulbs online, its predicted annual revenue will have increased by a total of approximately 518,748 dollars.
- C. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5% greater than the predicted annual revenue for the previous year.

$$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero



Which expression is equivalent to  $6x + 5x + 4y$ ?

- A.  $15x$
- B.  $15y$
- C.  $11x + 4y$
- D.  $30x + 4y$

If  $f(x) = \frac{x^2 - 6x + 3}{x - 1}$ , what is  $f(-1)$  ?

- A. -5
- B. -2
- C. 2
- D. 5

Q9

GRID-IN

ADVANCED MATH

$$x^2 = 2222$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which expression is equivalent to  $20w - 4w + 3w$ ?

A.  $10w$

B.  $13w$

C.  $19w$

D.  $21w$